INCH-POUND

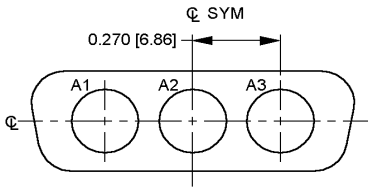
MIL-DTL-24308/31A
29 September 2022
SUPERSEDING
MIL-DTL-24308/31
w/AMENDMENT 1
20 November 2019

DETAIL SPECIFICATION SHEET

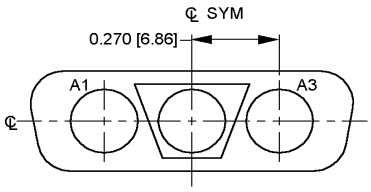
CONNECTORS, ELECTRIC, RECTANGULAR, MINIATURE, COMBINATION INSERT ARRANGEMENT, PLUG, PIN CONTACTS, SHELL SIZE 2, SIZE 20 SIGNAL CONTACTS AND SIZE 8 POWER AND SHIELDED, CLASS G, M, N AND D, PRINTED WIRING BOARD TYPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

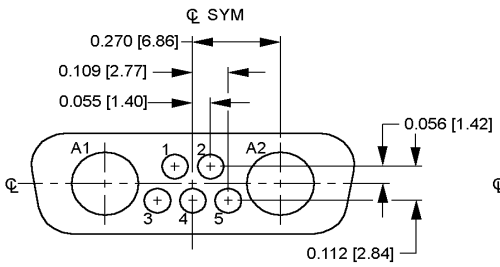
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24308.



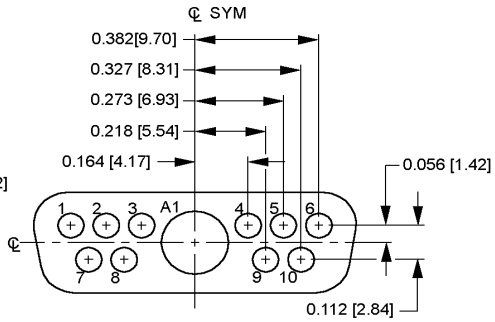
3W3



3WK3



7W2



11W1

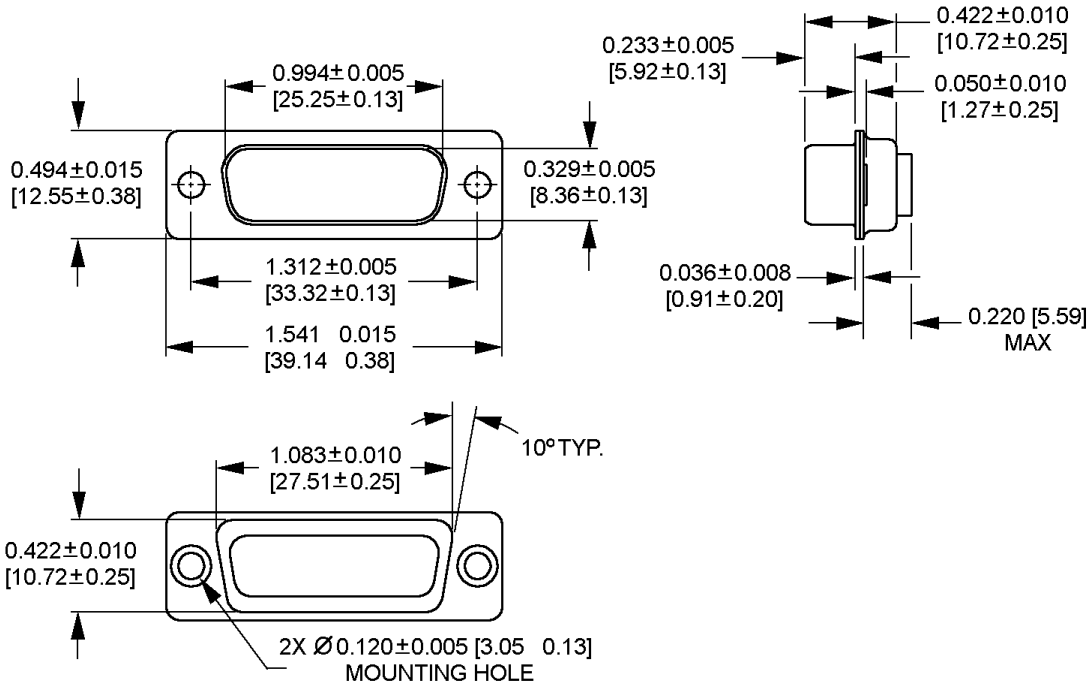
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).
4. All dimensions are true position (TP).
5. Contact holes in insert shall be located on TP within a .005 inch (0.13 mm) diameter circle.
6. Reference documents shall be of the issue in effect on date of invitations for bid.

FIGURE 1. Insert arrangements, shell size 2, pin contacts.

AMSC N/A
FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

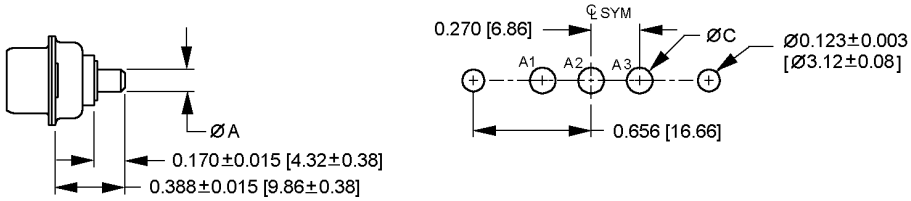
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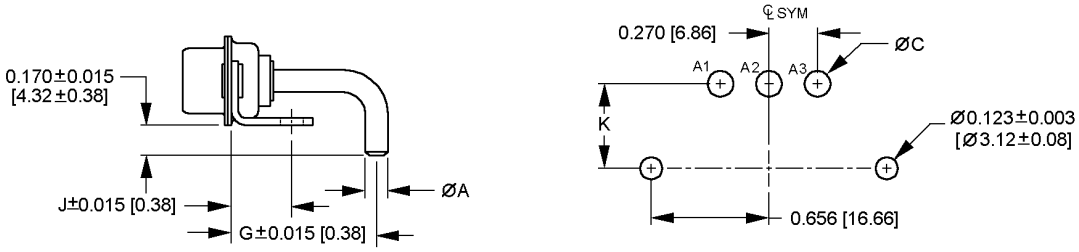
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).

FIGURE 2. Dimensions shell size 2, plug.

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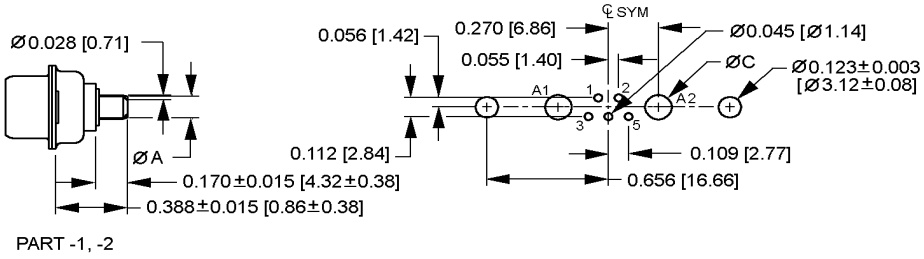


M24308/31-001, -002, -007, -008
M24308/31-101, -102, -107, -108



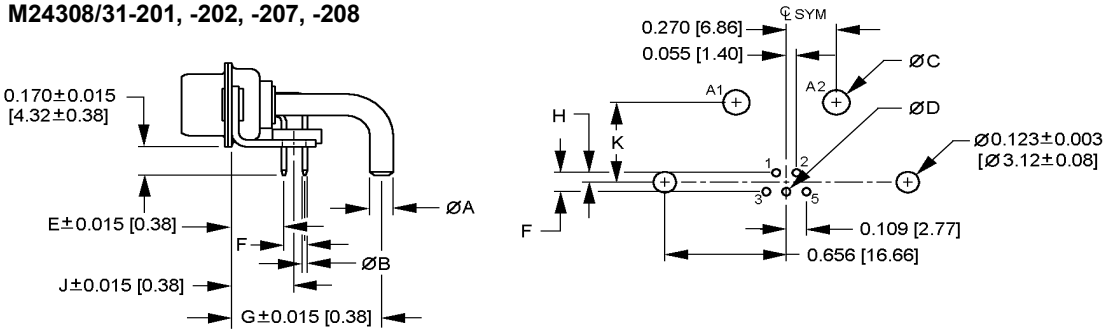
M24308/31-103, -104, 105, -106, -109, -110, -111, -112

3W3, 3WK3



PART -1, -2

M24308/31-201, -202, -207, -208

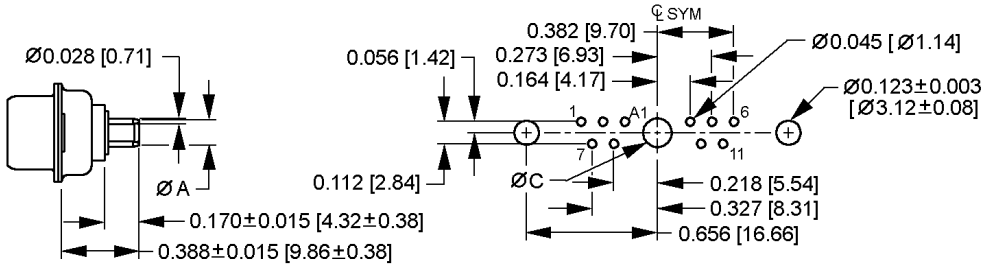


M24308/31-203, -204, -205, -206, -209, -210, -211, -212

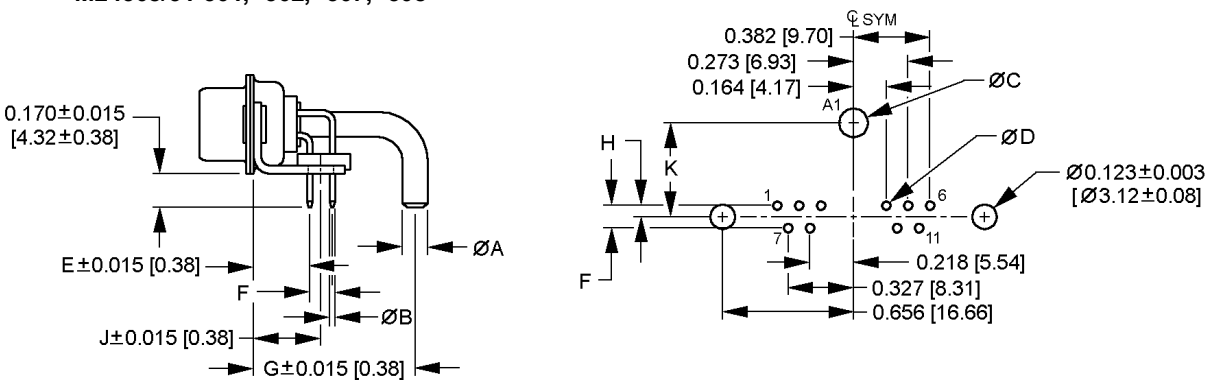
7W2

FIGURE 3. Termination and layout arrangements for power connectors, plug, printed wiring board type.

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Check the source to verify that this is the current version before use.



M24308/31-301, -302, -307, -308



M24308/31-303, -304, -305, -306, -309, -310, -311, -312

11W1

FIGURE 3. Termination and layout arrangements for power connectors, plug, printed wiring board type - Continued.

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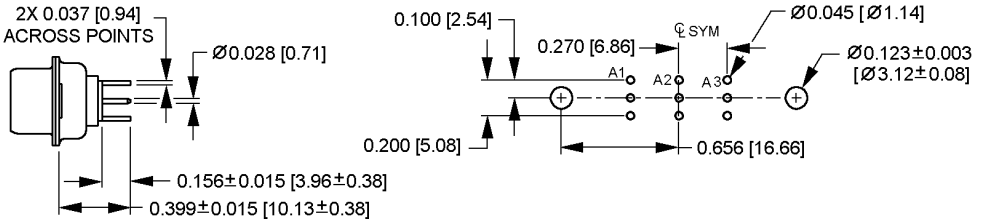
Dimensions and configurations.

PIN M24308/31	ØA	ØB	ØC	D	E	F	G	H	J	K
-001, -002 -101, -102	0.078 (1.98)	---	0.098 (2.49)	---	---	---	---	---	---	---
-007, -008 -107, -108	0.125 (3.18)	---	0.145 (3.68)	---	---	---	---	---	---	---
-003, -004 -113, -114	0.078 (1.98)	---	0.098 (2.49)	---	---	---	0.339 (8.61)	---	0.339 (8.61)	0.000 (0.00)
-009, -010 -109, -110	0.125 (3.18)	---	0.145 (3.68)	---	---	---	0.810 (20.57)	---	0.339 (8.61)	0.471 (11.96)
-005, -006 -105, -106	0.078 (1.98)	---	0.098 (2.49)	---	---	---	0.420 (10.67)	---	0.420 (10.67)	0.000 (0.00)
-011, -012 -111, -112	0.125 (3.18)	---	0.145 (3.68)	---	---	---	0.810 (20.57)	---	0.420 (10.67)	0.390 (9.91)
-201, -202 -301, -302	0.078 (1.98)	0.028 (0.71)	0.098 (2.49)	0.045 (1.14)	---	---	---	---	---	---
-207, -208 -307, -308	0.125 (3.18)	0.028 (0.71)	0.145 (3.68)	0.045 (1.14)	---	---	---	---	---	---
-203, -204 -303, -304	0.078 (1.98)	0.028 (0.71)	0.098 (2.49)	0.045 (1.14)	0.283 (7.19)	0.112 (2.84)	0.339 (8.61)	0.056 (1.42)	0.339 (8.61)	0.000 (0.00)
-209, -110 -309, -310	0.125 (3.18)	0.028 (0.71)	0.145 (3.68)	0.045 (1.14)	0.283 (7.19)	0.112 (2.84)	0.810 (20.57)	0.056 (1.42)	0.339 (8.61)	0.471 (11.96)
-205, -206 -305, -306	0.078 (1.98)	0.024 (0.61)	0.098 (2.49)	0.040 (1.02)	0.370 (9.40)	0.100 (2.54)	0.420 (10.67)	0.050 (1.27)	0.420 (10.67)	0.000 (0.00)
-211, -212 -311, -312	0.125 (3.18)	0.024 (0.61)	0.145 (3.68)	0.040 (1.02)	0.370 (9.40)	0.100 (2.54)	0.810 (20.57)	0.050 (1.27)	0.420 (10.67)	0.390 (9.91)

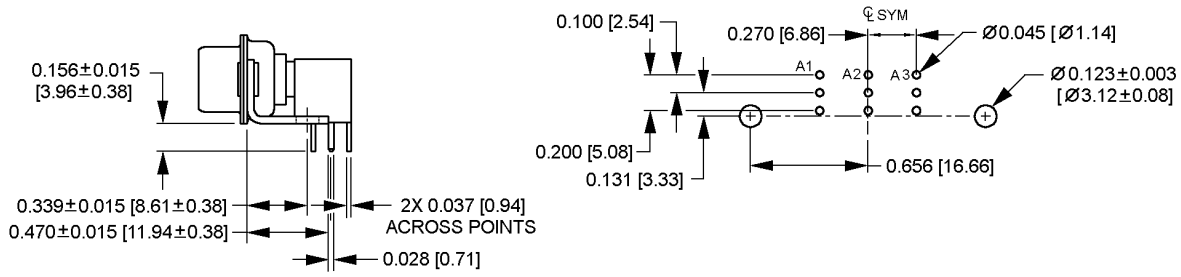
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).

FIGURE 3. Termination and layout arrangements for power connectors, plug, printed wiring board type. - Continued.

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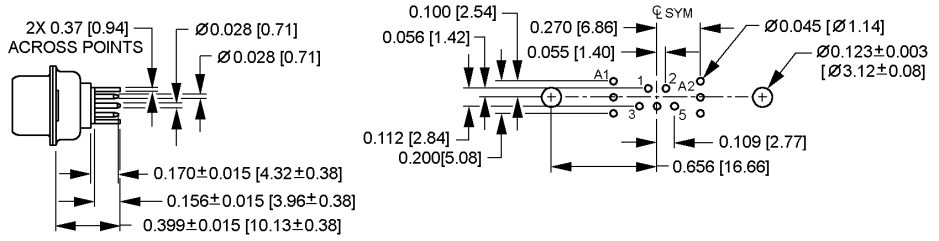


M24308/31-013, -014
M24308/31-113, -114

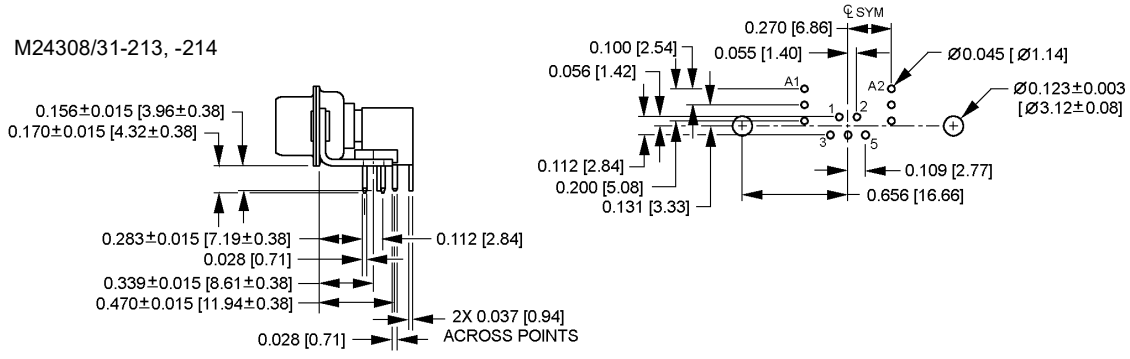


M24308/31-015, -016
M24308/31-115, -116

3W3 AND 3WK3



M24308/31-213, -214

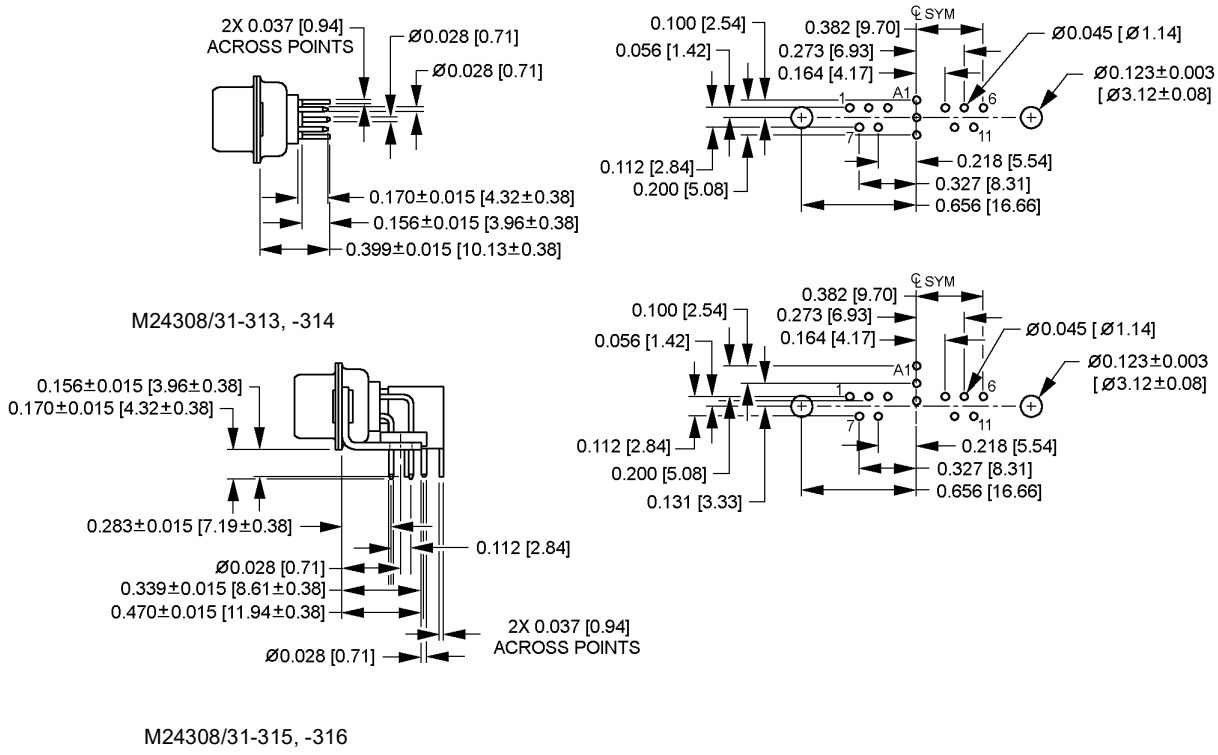


M24308/31-215, -216

7W2

FIGURE 4. Termination and layout arrangements for shielded connectors, plug, printed wiring board type.

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11W1

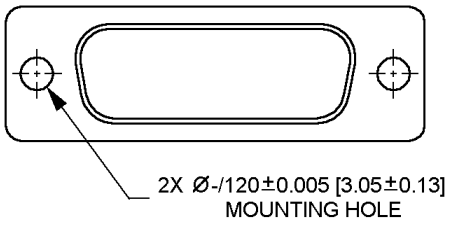
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).

FIGURE 4. Termination and layout arrangements for shielded connectors, plug, printed wiring board type - Continued.

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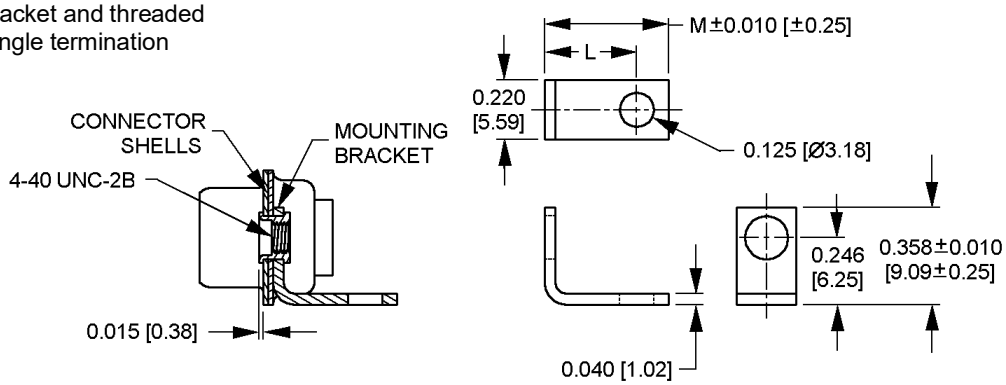
CONFIGURATION A:

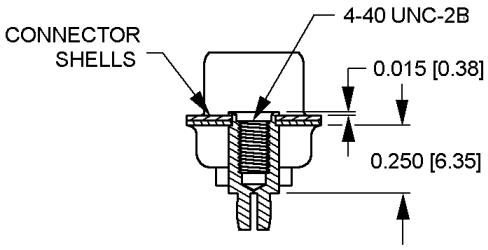
Ø .120 (3.05 mm) mounting hole, straight termination connectors



CONFIGURATION B AND C:

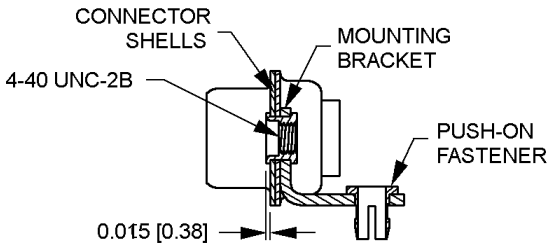
Mounting bracket and threaded rivet, right angle termination connectors





CONFIGURATION D:

Swaged spacer with push on fastener, straight termination connectors



CONFIGURATION E AND F:

Mounting bracket and threaded rivet with push on fastener, right angle termination connectors

FIGURE 5. Mounting configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:15Z
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Mounting bracket configuration	L	M	Mounting type
B	0.339	0.456	0.125 hole
C	0.420	0.537	0.125 hole
E	0.339	0.456	Fastener
F	0.420	0.537	Fastener

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).

FIGURE 5. Mounting configurations - Continued.

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REQUIREMENTS:

Dimensions and configurations:

Insert arrangements shall be in accordance with figure 1.

Shell configuration shall be in accordance with figure 2.

Termination and layout arrangement shall be in accordance with figures 3 and 4

Mounting configurations shall be in accordance with figure 5.

Materials and finishes:

Mounting brackets half hard brass, UNS-C26000 in accordance with ASTM B36/B36M or corrosion-resistant steel shall be passivated in accordance with SAE AMS2700 or ASTM A580/A580M.

Finish in accordance with MIL-DTL-24308.

Spacer with push fastener, phosphor bronze, UNS-C54400 in accordance with ASTM B139/B139M.

Finish: Gold in accordance with MIL-DTL-45204, type II, grade C, class 1 or equivalent over copper in accordance with SAE AMS 2418.

Push fastener with mounting brackets, beryllium copper, UNS-C17300 in accordance with ASTM B196/B196M.

Finish: Gold in accordance with MIL-DTL-45204, type II, grade C, class 1 or equivalent over copper in accordance with SAE AMS 2418.

Electrical:

Size 20 signal contacts:

Current rating:	7.5 amps nominal
Initial contact resistance:	.008 ohms maximum
Proof voltage:	1000 V r.m.s.

Power contacts:

.078 dia:	10 amps
.125 dia:	40 amps
Initial contact resistance:	.0005 ohms maximum

Shielded contacts:

Nominal impedance:50 ohms

Insertion loss:	-0.46 dB at 1 GHz
	-1.5 dB at 2 GHz
VSWR:	1.15 average at 1 GHz
	1.56 average at 2 GHz
Proof voltage:	1000 V r.m.s
Initial contact resistance:	.008 ohms maximum

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TABLE I. PIN and dash numbers. 1/

PIN	Insert arrangement (see figure 1)	Size 8 contact	Termination type	Mounting configuration (see figure 5)
M24308/31-001	3WK3	0.078	Straight	A
M24308/31-002		0.078	Straight	D
M24308/31-003		0.078	Right angle	B
M24308/31-004		0.078	Right angle	E
M24308/31-005		0.078	Right angle	C
M24308/31-006		0.078	Right angle	F
M24308/31-007		0.125	Straight	A
M24308/31-008		0.125	Straight	D
M24308/31-009		0.125	Right angle	B
M24308/31-010		0.125	Right angle	E
M24308/31-011		0.125	Right angle	C
M24308/31-012		0.125	Right angle	F
M24308/31-013		Shielded	Straight	A
M24308/31-014		Shielded	Straight	D
M24308/31-015		Shielded	Right angle	B
M24308/31-016		Shielded	Right angle	E
M24308/31-101	3W3	0.078	Straight	A
M24308/31-102		0.078	Straight	D
M24308/31-103		0.078	Right angle	B
M24308/31-104		0.078	Right angle	E
M24308/31-105		0.078	Right angle	C
M24308/31-106		0.078	Right angle	F
M24308/31-107		0.125	Straight	A
M24308/31-108		0.125	Straight	D
M24308/31-109		0.125	Right angle	B
M24308/31-110		0.125	Right angle	E
M24308/31-111		0.125	Right angle	C
M24308/31-112		0.125	Right angle	F
M24308/31-113		Shielded	Straight	A
M24308/31-114		Shielded	Straight	D
M24308/31-115		Shielded	Right angle	B
M24308/31-116		Shielded	Right angle	E

1/ The connector mates with MIL-DTL-24308/32.

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TABLE I. PIN and dash numbers - Continued. 1/

PIN	Insert arrangement (see figure 1)	Size 8 contact	Termination type	Mounting configuration (see figure 5)
M24308/31-201	7W2	0.078	Straight	A
M24308/31-202		0.078	Straight	D
M24308/31-203		0.078	Right angle	B
M24308/31-204		0.078	Right angle	E
M24308/31-205		0.078	Right angle	C
M24308/31-206		0.078	Right angle	F
M24308/31-207		0.125	Straight	A
M24308/31-208		0.125	Straight	D
M24308/31-209		0.125	Right angle	B
M24308/31-210		0.125	Right angle	E
M24308/31-211		0.125	Right angle	C
M24308/31-212		0.125	Right angle	F
M24308/31-213		Shielded	Straight	A
M24308/31-214		Shielded	Straight	D
M24308/31-215		Shielded	Right angle	B
M24308/31-216		Shielded	Right angle	E
M24308/31-301	11W1	0.078	Straight	A
M24308/31-302		0.078	Straight	D
M24308/31-303		0.078	Right angle	B
M24308/31-304		0.078	Right angle	E
M24308/31-305		0.078	Right angle	C
M24308/31-306		0.078	Right angle	F
M24308/31-307		0.125	Straight	A
M24308/31-308		0.125	Straight	D
M24308/31-309		0.125	Right angle	B
M24308/31-310		0.125	Right angle	E
M24308/31-311		0.125	Right angle	C
M24308/31-312		0.125	Right angle	F
M24308/31-313		Shielded	Straight	A
M24308/31-314		Shielded	Straight	D
M24308/31-315		Shielded	Right angle	B
M24308/31-316		Shielded	Right angle	E

1/ The connector mates with MIL-DTL-24308/32.

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PIN example: M24308/31 -105 F G

Specification sheet number:

Dash number from table I:

Finish:

A = Pure electrodeposited aluminum. 1/
F = Cadmium. 1/, 5/
K = Zinc nickel. 1/ 2/
P = Passivated stainless steel. 1/
T = Nickel Fluorocarbon Polymer. 1/ 2/
Z = zinc. 1/
- = nickel 3/ or gold 4/

Class:

G, N, D or M (See MIL-DTL-24308)

NOTES:
1/ Finish for class G or N.
2/ Not for use with class N or M
3/ Finish for class D.
4/ Finish for class M.
5/ Cadmium finish may have a chromate conversion coating or chromic acid rinse which includes hexavalent chromium and are not recommended for Army ground systems.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Referenced documents. In addition to MIL-DTL-24308, this document references the following:

MIL-DTL-24308/32
MIL-DTL-45204
ASTM B36/B36M
ASTM B139/B139M
ASTM B196/B196M
ASTM A580/A580M
SAE AMS2700
SAE AMS 2418

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force – 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project: 5935-2022-150)

Review activities:
Army - AT, CR4, MI
Navy - AS, CG, MC, SH
MISC - MDA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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